

Traffic Flow Analysis Report

Video-Based Vehicle Classification & Level of Service (LOS) Assessment

Analyzed Clip: 5.10 minutes (9,184 frames @ 30 fps) | Assumed Lane Capacity: 1,500 PCU/hr

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1. Executive Summary

A 5.10-minute video clip was processed using an automated vehicle detection and classification pipeline to estimate traffic volume and Level of Service (LOS) across four observed lanes. A total of 267 vehicles were detected and converted to 455.0 Passenger Car Units (PCU) using standard PCU equivalency factors.

Three of the four lanes are operating at or beyond their assumed capacity of 1,500 PCU/hr. Lane 3 shows the most severe congestion ($V/C = 1.239$, LOS F), driven by a heavy mix of trucks and buses despite carrying the fewest vehicles of any lane. Lane 1 is also in breakdown condition (LOS F) and Lane 2 is operating right at capacity (LOS E). Lane 4 is the only lane with comfortable spare capacity (LOS B).

267	455.0	3 of 4
Total Vehicles Detected	Total PCU	Lanes at/over Capacity (E or F)

2. Methodology

Vehicles in the video clip were detected and classified frame-by-frame into five categories (car, truck, bus, motorcycle, bicycle). Each class was converted to Passenger Car Units (PCU) using standard equivalency factors, and the accumulated PCU over the clip duration was extrapolated to an hourly flow rate (PCU/hr).

The flow rate for each lane was then divided by an assumed lane capacity of 1,500 PCU/hr to obtain a volume-to-capacity (V/C) ratio. This ratio was mapped to a standard Level of Service (LOS) grade (A = free flow through F = breakdown flow), following the general approach of the HCM / IRC level-of-service framework.

LOS Grade Reference

Grade	V/C Range	Description
A	0.00 - 0.20	Free flow, minimal delay
B	0.21 - 0.44	Reasonably free flow
C	0.45 - 0.74	Stable flow, acceptable delay
D	0.75 - 0.84	Approaching unstable flow
E	0.85 - 1.00	Unstable flow / at capacity
F	> 1.00	Forced / breakdown flow

Note: results depend on the analyzed clip length and the assumed design capacity; they should be treated as indicative rather than an official traffic engineering assessment.

2.1 Detection Model & Software Stack

Vehicle detection and classification were performed using YOLOv8n (“nano”), the smallest and fastest variant in Ultralytics’ YOLOv8 family, chosen for its strong speed/accuracy trade-off on resource-constrained hardware. The pretrained model file used is yolov8n.pt (COCO-pretrained weights, ~6.4 MB / 3.2M parameters).

- Model: YOLOv8n (Ultralytics YOLOv8, nano variant)
- Reported accuracy (official Ultralytics COCO benchmark): mAP@0.5:0.95 = 37.3% on the COCO val dataset, at ~0.99 ms inference per image on an NVIDIA A100 (TensorRT). This is the published benchmark for the stock yolov8n.pt weights and is not a measurement specific to this clip.
- Core libraries: Ultralytics (YOLOv8 implementation), PyTorch (model inference backend), OpenCV (video I/O and frame processing), NumPy/Pandas (data aggregation), Matplotlib (chart generation).
- Language & runtime: Python 3, organized as a multi-module pipeline (main.py, config.py, tracker_utils.py, calibrate_lanes.py, generate_report.py).

3. Lane-Wise Results

Lane	Car	Truck	Bus	Total Veh.	Total PCU	Flow (PCU/hr)	V/C	LOS
1	53	17	10	80	134.0	1575.78	1.051	F
2	87	4	9	100	126.0	1481.71	0.988	E
3	5	26	25	56	158.0	1858.01	1.239	F
4	28	0	3	31	37.0	435.10	0.290	B

Grand Total	267 vehicles	455.0 PCU
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4. Visual Summary

4.1 Vehicle Volume by Lane

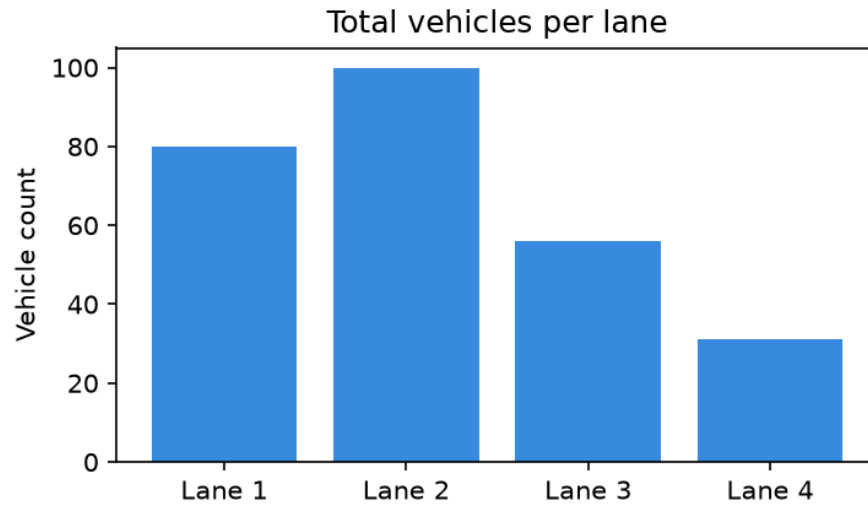


Figure 1: Total vehicle count per lane. Lane 2 carries the most vehicles (100), Lane 4 the fewest (31).

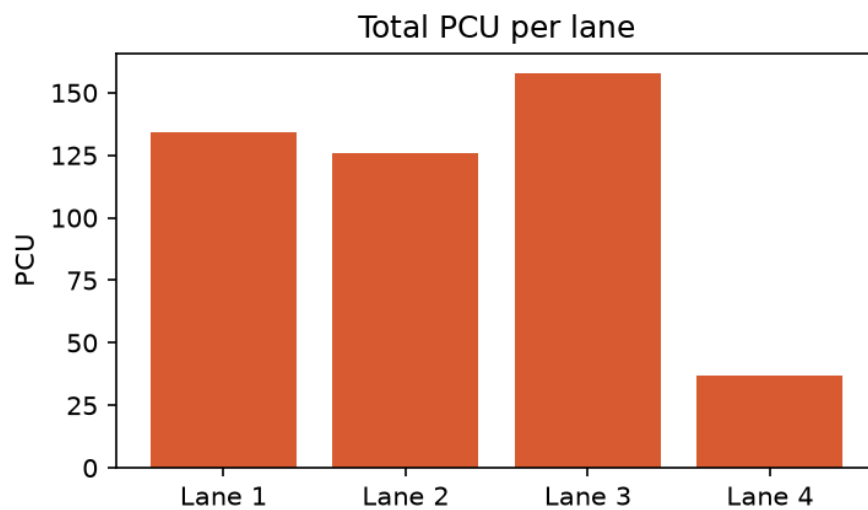


Figure 2: Total PCU per lane. Lane 3 has the highest PCU load (158.0) despite having the second-fewest vehicles, reflecting its heavy truck/bus mix.

4.2 Congestion (V/C Ratio & LOS)

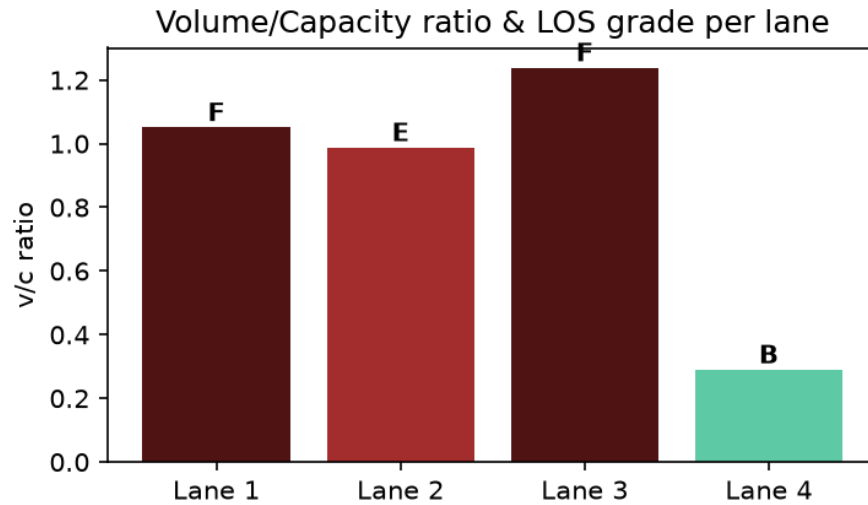


Figure 3: V/C ratio and LOS grade per lane. Lanes 1 and 3 exceed capacity (LOS F); Lane 2 is at capacity (LOS E); Lane 4 has ample spare capacity (LOS B).

4.3 Overall Vehicle Composition

Overall vehicle type distribution

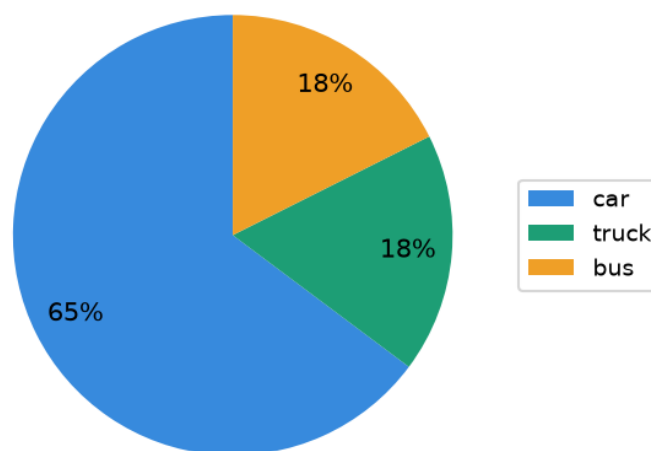


Figure 4: Aggregate vehicle mix across all lanes - 65% cars, 18% trucks, 18% buses. No motorcycles or bicycles were detected.

5. Discussion

- Lane 3 - critical congestion driven by vehicle mix: With only 56 vehicles (the second-lowest count), Lane 3 still produces the highest PCU load (158.0) and the worst V/C ratio (1.239). This is because 91% of its traffic is trucks and buses, which carry much higher

PCU weightings than cars. This lane behaves as a heavy/commercial-vehicle corridor and is the most over capacity of all four lanes.

- Lane 1 - high volume and a non-trivial heavy-vehicle share: Lane 1 has the highest vehicle-to-PCU conversion impact among the higher-volume lanes, with roughly 34% of its 80 vehicles being trucks or buses. Combined with its high raw vehicle count, this pushes it into LOS F ($V/C = 1.051$).
- Lane 2 - high volume, lighter mix, right at the edge: Lane 2 carries the most vehicles of any lane (100) but is dominated by cars (87%), keeping its PCU load comparatively lower (126.0). It sits just under capacity at LOS E ($V/C = 0.988$) - effectively the threshold between stable and breakdown flow.
- Lane 4 - the outlier with spare capacity: Lane 4 has the fewest vehicles (31), a car-dominated mix, and no trucks at all in this sample. Its V/C ratio of 0.29 (LOS B) indicates substantial unused capacity relative to the other three lanes.
- Vehicle composition overall: Across all lanes combined, the mix is 65% cars, 18% trucks, and 18% buses, with no motorcycles or bicycles detected in this clip. The truck/bus share is disproportionately concentrated in Lanes 1 and 3, which is the primary driver of those lanes' poor LOS grades.

6. Conclusions & Recommendations

- Three of four lanes (1, 2, 3) are operating at or beyond the assumed 1,500 PCU/hr capacity, indicating the intersection/road segment as a whole is under significant pressure during the observed period.
- Lane 3 should be flagged as the highest priority for intervention - its breakdown condition is driven by heavy-vehicle concentration rather than sheer vehicle count, suggesting measures such as a dedicated heavy-vehicle lane, signal timing adjustments, or restricting truck/bus movements during peak periods.
- Lane 2's near-capacity status ($V/C = 0.988$) means it is highly sensitive to small increases in volume; even modest growth could tip it into LOS F.
- Lane 4's spare capacity suggests it may be under-utilized; if feasible, channelizing some traffic from Lanes 1-3 toward Lane 4 (via lane markings, signage, or signal phasing) could help redistribute load.
- Because this assessment is based on a 5.10-minute clip and an assumed capacity value, it should be treated as indicative. A longer observation window (ideally covering peak and off-peak periods) and a capacity figure calibrated to the road's actual design standard are recommended before using these findings for formal planning decisions.